PRACTICAL APPLICATION – 1
Threat Hunting Practice

1. Objective
To perform threat hunting using a combination of Elastic Security, AlienVault OTX, and Velociraptor in order to identify suspicious activities, validate indicators of compromise (IOCs), and map findings to MITRE ATT&CK framework.

2. Tools Used
- Elastic Security (SIEM Log Analysis)
- AlienVault OTX (Threat Intelligence Platform)
- Velociraptor (Endpoint Forensics – Pslist Artifact)

3. Hypothesis Development
A threat hunting hypothesis was formulated as follows:
"Unauthorized privilege escalation activity may be occurring in domain accounts through abnormal assignment of administrative privileges."
This hypothesis was validated by querying security logs for Event ID 4672, which indicates special privilege assignment.

4. Elastic Security Analysis
A query was executed in Elastic Security to identify Event ID **4672 (Special Privileges Assigned)**.
Findings:

Timestamp	User	Event ID	Notes
2026-04-24 15:30:00	testuser	4672	Unexpected admin role

Observation:
The log indicates a potential unauthorized elevation of privileges, which is a strong indicator of suspicious administrative activity.

5. Threat Intelligence Analysis (AlienVault OTX)
AlienVault OTX was used to investigate **MITRE ATT&CK Technique T1078 (Valid Accounts)**.
Findings:
- Multiple OTX pulses identified ("Oz Batch: 50 IOCs")
- Average Behavioral Detection Score (BDE): ~80
- IP addresses originating from multiple regions including USA, Brazil, Asia, and Europe
- Indicators suggest possible reconnaissance and initial access activity

Conclusion:
The indicators align with potential misuse of valid credentials and reconnaissance activity.

6. Velociraptor Endpoint Analysis
Velociraptor was deployed on an Ubuntu endpoint using the **Windows.System.Pslist artifact** to perform process enumeration.

Execution Summary:
- Server successfully initialized and GUI accessed via https://localhost:8889
- Client successfully enrolled (Client ID: C.a4e71d59e43edbd3)
- Hunt created and executed (H.D7LI5U7VOB4AC)
- Artifact executed successfully on endpoint

Evidence Collected:
- Hunt metadata (hunt_info.json)
- Client logs (log.json, requests.json)
- Collection context files

Observation:
The exported Pslist dataset did not return active process entries at execution time, but successful client communication and hunt execution were verified.

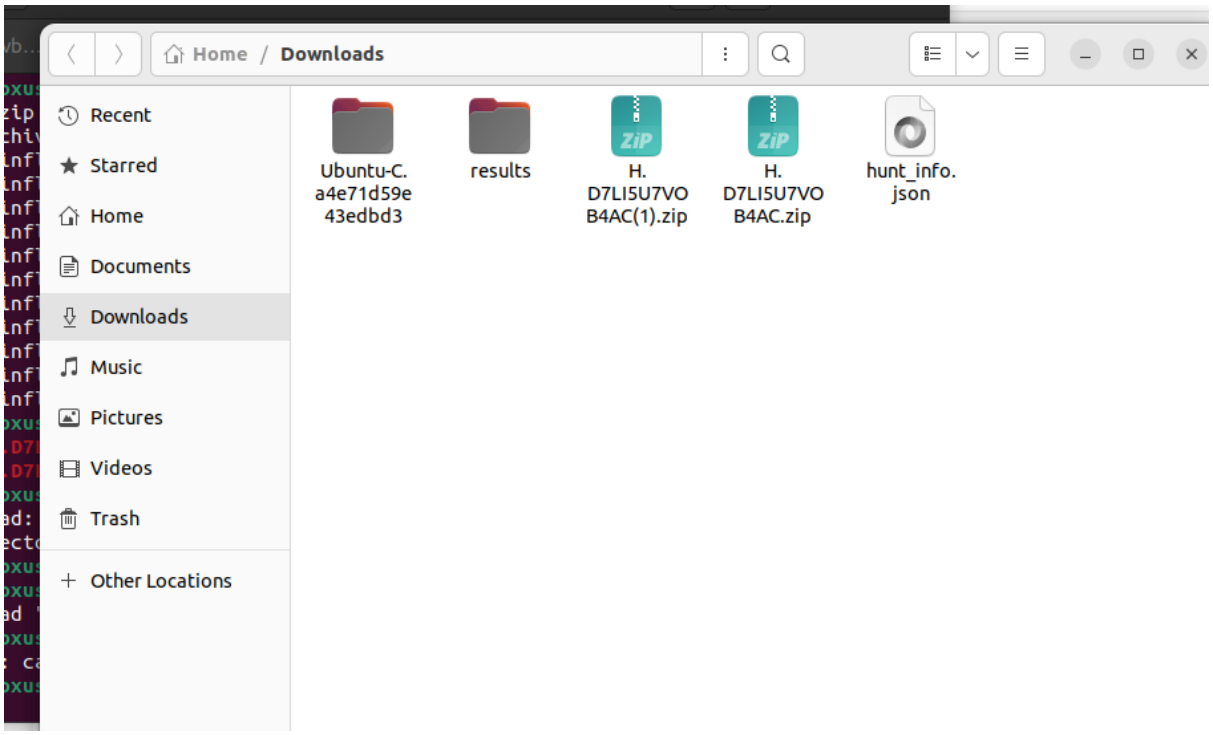
7. MITRE ATT&CK Mapping

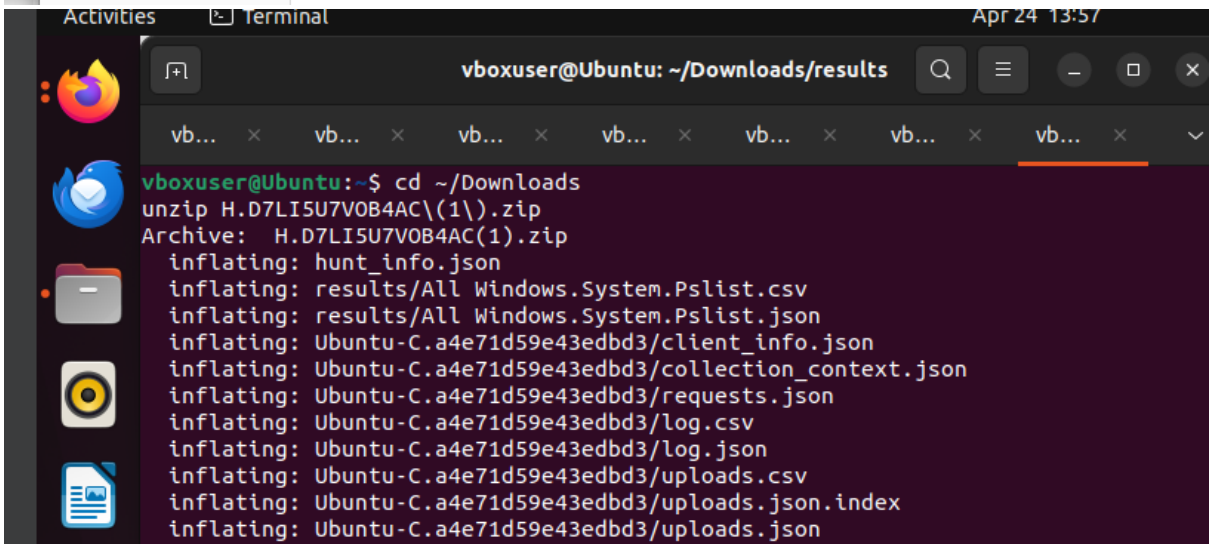
Technique ID	Description
T1078	Valid Accounts (Unauthorized privilege escalation and misuse of credentials)

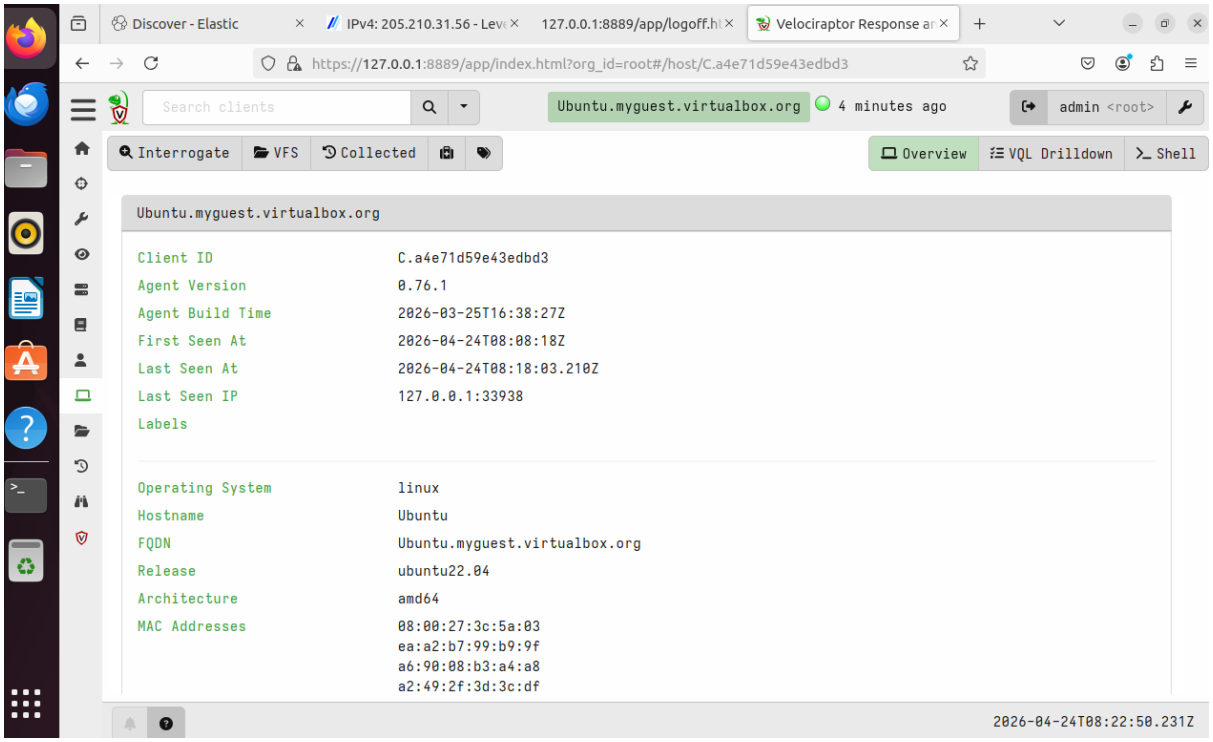
8. Hunting Report Summary (100 Words)
Threat hunting was performed using Elastic Security, AlienVault OTX, and Velociraptor. A suspicious privilege escalation event (Event ID 4672) was identified in Elastic logs, indicating potential unauthorized administrative access. AlienVault OTX analysis of MITRE ATT&CK T1078 revealed multiple high-risk IP indicators associated with reconnaissance and possible initial access activity. Velociraptor was deployed for endpoint investigation using the Windows.System.Pslist artifact. Although no active process data was returned, successful client connectivity and hunt execution were confirmed. The exercise demonstrates an effective multi-layered threat hunting approach combining SIEM, threat intelligence, and endpoint forensic analysis to identify and validate potential security threats.

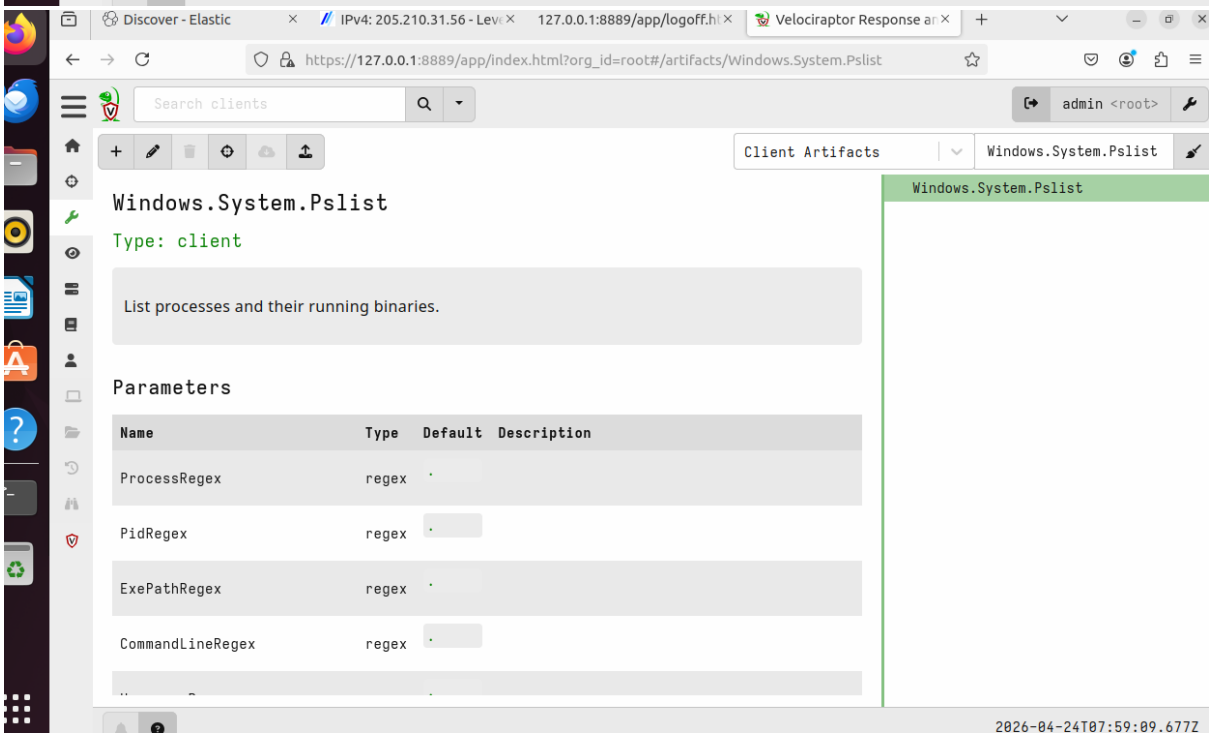
9. Conclusion
The threat hunting exercise successfully demonstrated the integration of SIEM, threat intelligence, and endpoint forensic tools. Elastic Security identified suspicious privilege escalation activity, AlienVault OTX provided contextual threat intelligence aligned with MITRE ATT&CK T1078, and Velociraptor confirmed endpoint visibility and execution capability. This exercise highlights the importance of correlating multiple security data sources to improve detection accuracy and incident response readiness.

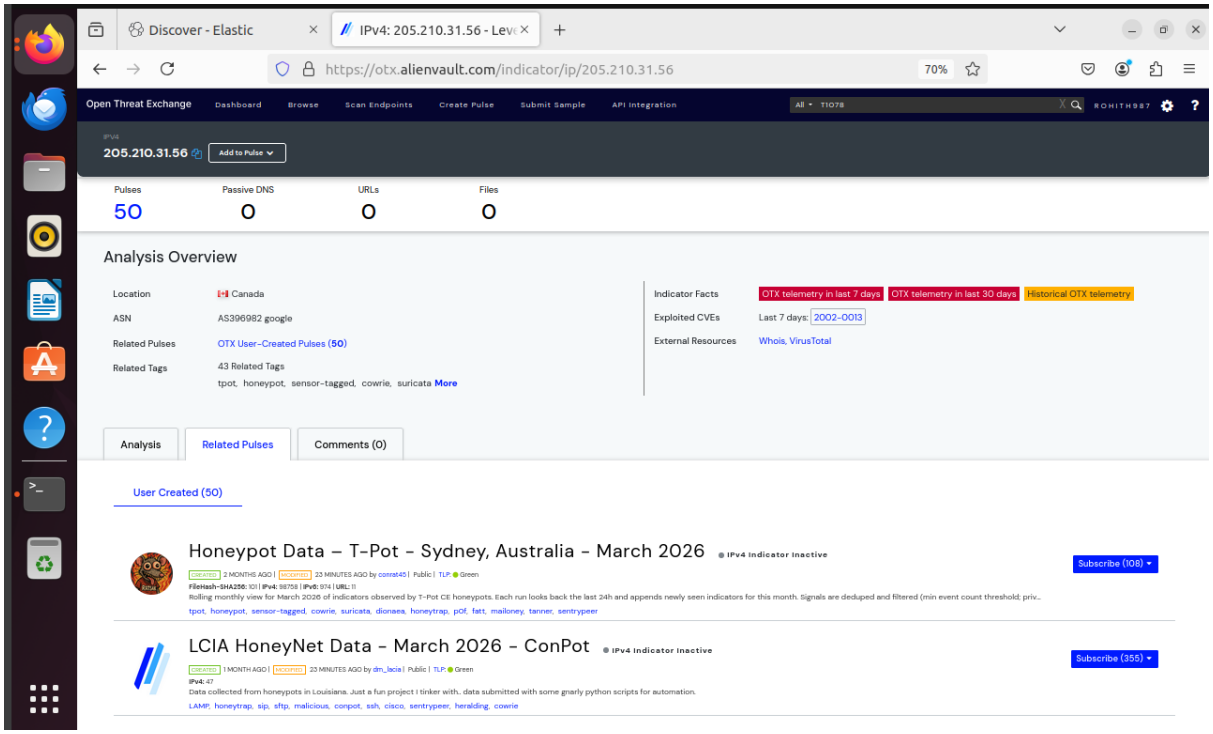
Screenshots :

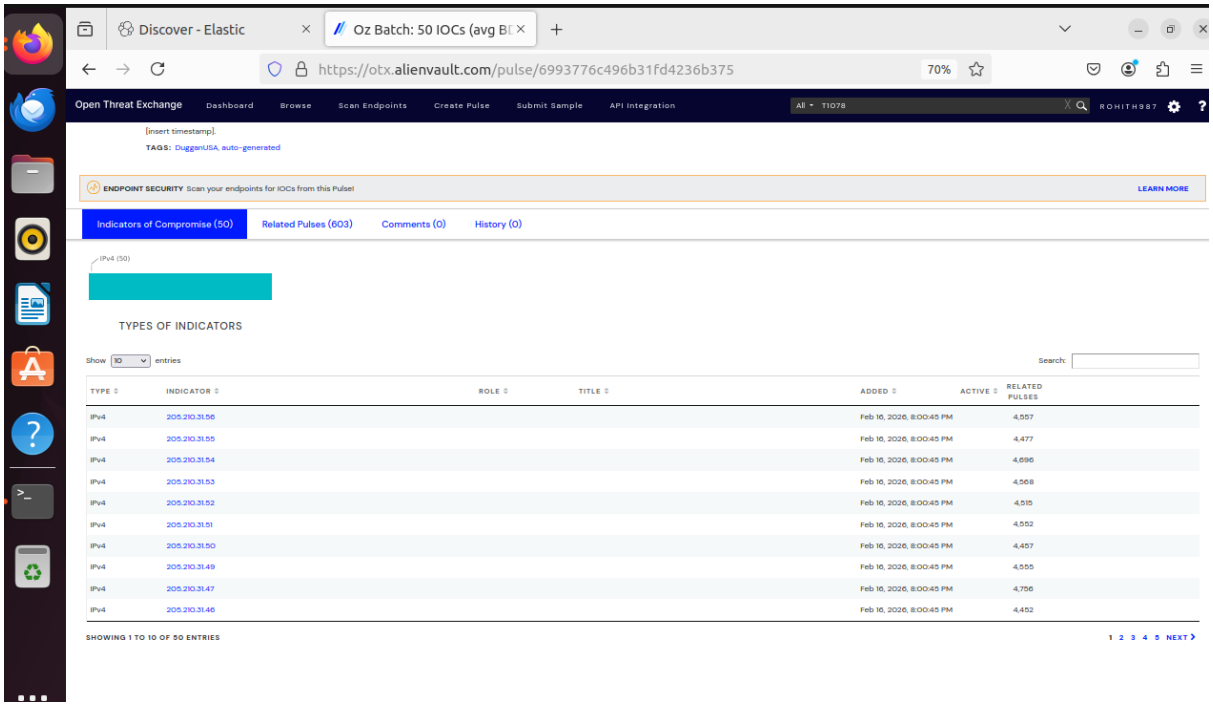


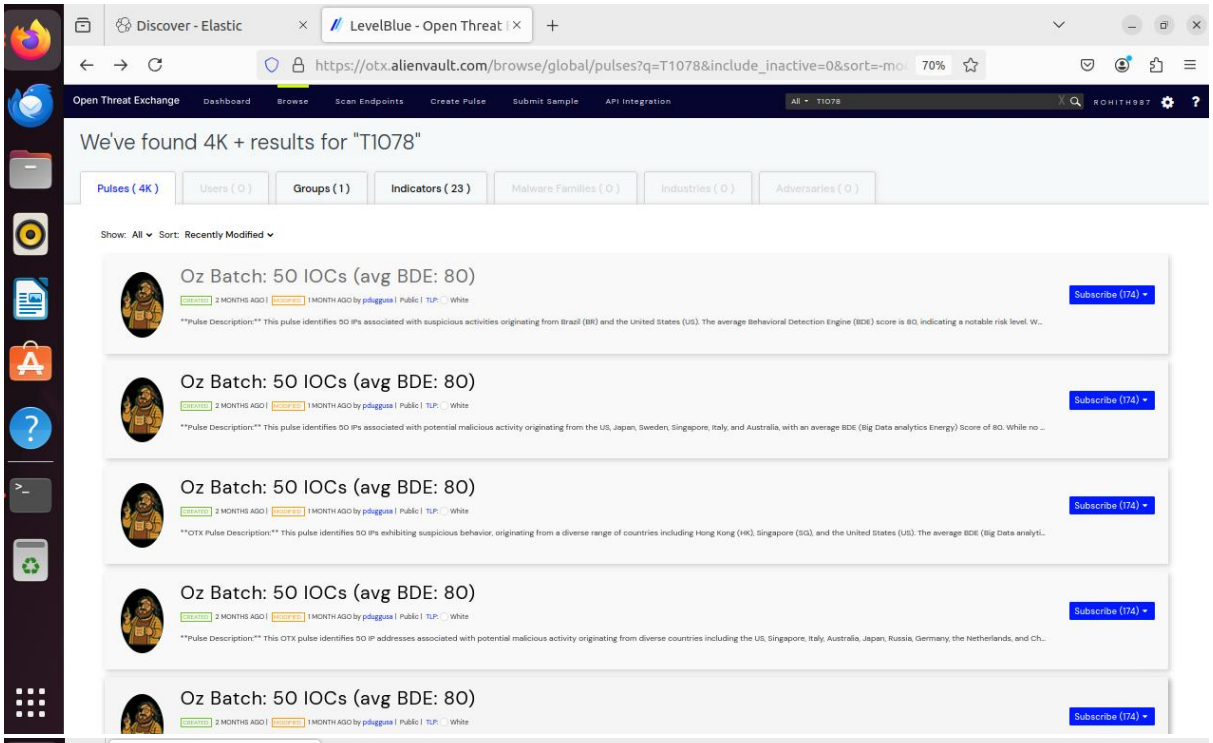



Discover - Elastic
IPv4: 205.210.31.56 - Leve
127.0.0.1:8889/app/logoff.ht
Velociraptor Response ar
https://127.0.0.1:8889/app/index.html?org_id=root#/host/C.a4e71d59e43edbd3
Search clients
Ubuntu.myguest.virtualbox.org
4 minutes ago
admin <root>
Interrogate
VFS
Collected
Overview
VQL Drilldown
Shell
Ubuntu.myguest.virtualbox.org
Client ID C.a4e71d59e43edbd3
Agent Version 0.76.1
Agent Build Time 2026-03-25T16:38:27Z
First Seen At 2026-04-24T08:08:18Z
Last Seen At 2026-04-24T08:18:03.210Z
Last Seen IP 127.0.0.1:33938
Labels
Operating System linux
Hostname Ubuntu
FQDN Ubuntu.myguest.virtualbox.org
Release ubuntu22.04
Architecture amd64
MAC Addresses 08:00:27:3c:5a:03
ea:a2:b7:99:b9:9f
a6:90:08:b3:a4:a8
a2:49:2f:3d:3c:df
2026-04-24T08:22:50.231Z


Discover - Elastic
IPv4: 205.210.31.56 - Leve
127.0.0.1:8889/app/logoff.ht
Velociraptor Response ar
https://127.0.0.1:8889/app/index.html?org_id=root#/artifacts/Windows.System.Pslist
Search clients
admin <root>
Client Artifacts
Windows.System.Pslist
Windows.System.Pslist
Type: client
List processes and their running binaries.
Parameters
Name Type Default Description
ProcessRegex regex
PidRegex regex
ExePathRegex regex
CommandLineRegex regex
2026-04-24T07:59:09.677Z


Discover - Elastic
IPv4: 205.210.31.56 - Lev
https://otx.alienvault.com/indicator/ip/205.210.31.56
70%
Open Threat Exchange
Dashboard
Browse
Scan Endpoints
Create Pulse
Submit Sample
API Integration
ROHITH987
205.210.31.56
Add to Pulse
Pulses
50
Passive DNS
0
URLs
0
Files
0
Analysis Overview
Location
Canada
ASN
AS396982 google
Related Pulses
OTX User-Created Pulses (50)
Related Tags
43 Related Tags
tpot, honeypot, sensor-tagged, cowrie, suricata More
Indicator Facts
OTX telemetry in last 7 days
OTX telemetry in last 30 days
Historical OTX telemetry
Exploited CVEs
Last 7 days: 2002-0013
External Resources
Whois, VirusTotal
Analysis
Related Pulses
Comments (0)
User Created (50)
Honeypot Data – T-Pot - Sydney, Australia - March 2026
IPv4 Indicator Inactive
Subscribe (108)
LCIA HoneyNet Data - March 2026 - ConPot
IPv4 Indicator Inactive
Subscribe (355)


Discover - Elastic
Oz Batch: 50 IOCs (avg B
https://otx.alienvault.com/pulse/6993776c496b31fd4236b375
70%
Open Threat Exchange
Dashboard
Browse
Scan Endpoints
Create Pulse
Submit Sample
API Integration
ROHITH987
TAGS: DugganUSA, auto-generated
ENDPOINT SECURITY Scan your endpoints for IOCs from this Pulse!
LEARN MORE
Indicators of Compromise (50)
Related Pulses (603)
Comments (0)
History (0)
TYPES OF INDICATORS
Show 10 entries
Search:
TYPE
INDICATOR
ROLE
TITLE
ADDED
ACTIVE
RELATED PULSES
IPv4 205.210.31.56 Feb 16, 2026, 8:00:45 PM 4,557
IPv4 205.210.31.55 Feb 16, 2026, 8:00:45 PM 4,477
IPv4 205.210.31.54 Feb 16, 2026, 8:00:45 PM 4,696
IPv4 205.210.31.53 Feb 16, 2026, 8:00:45 PM 4,568
IPv4 205.210.31.52 Feb 16, 2026, 8:00:45 PM 4,515
IPv4 205.210.31.51 Feb 16, 2026, 8:00:45 PM 4,552
IPv4 205.210.31.50 Feb 16, 2026, 8:00:45 PM 4,457
IPv4 205.210.31.49 Feb 16, 2026, 8:00:45 PM 4,555
IPv4 205.210.31.47 Feb 16, 2026, 8:00:45 PM 4,756
IPv4 205.210.31.46 Feb 16, 2026, 8:00:45 PM 4,452
SHOWING 1 TO 10 OF 50 ENTRIES
1 2 3 4 5 NEXT


Discover - Elastic
LevelBlue - Open Threat
https://otx.alienvault.com/browse/global/pulses?q=T1078&include_inactive=0&sort=-mo
Open Threat Exchange
Dashboard
Browse
Scan Endpoints
Create Pulse
Submit Sample
API Integration
We've found 4K + results for "T1078"
Pulses (4K)
Groups (1)
Indicators (23)
Show: All
Sort: Recently Modified
Oz Batch: 50 IOCs (avg BDE: 80)
Subscribe (174)
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Oz Batch: 50 IOCs (avg BDE: 80)
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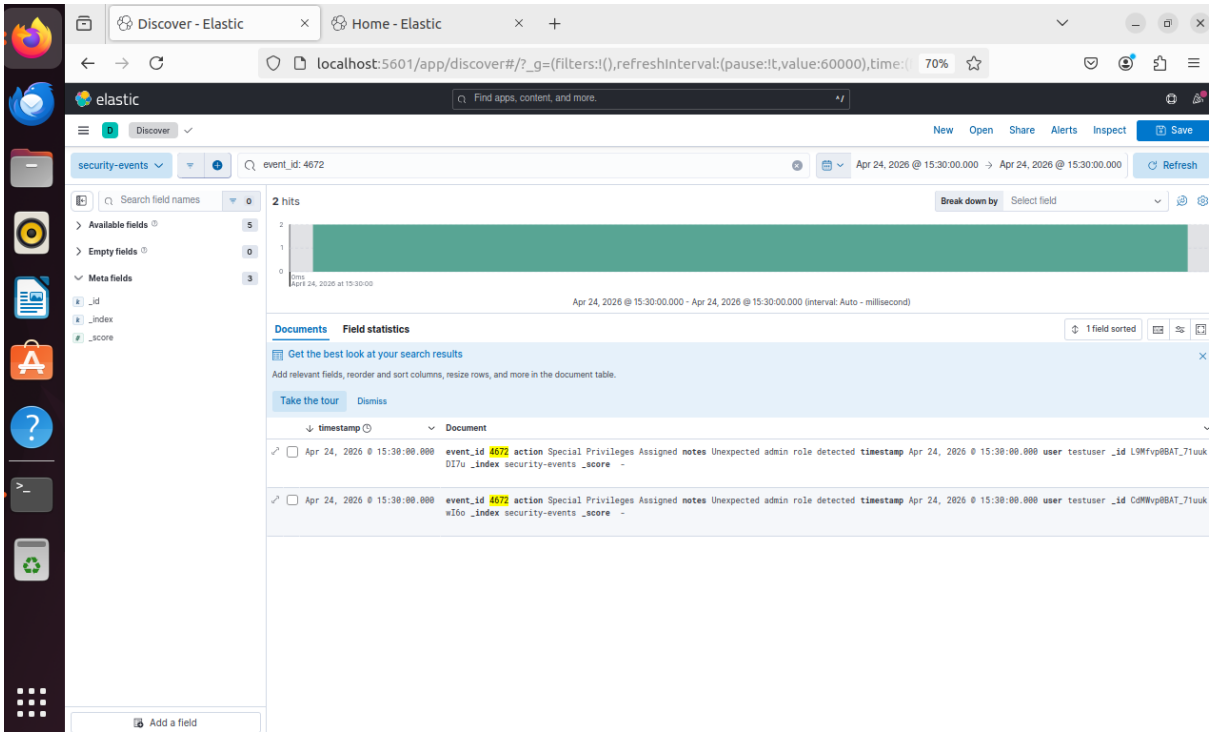
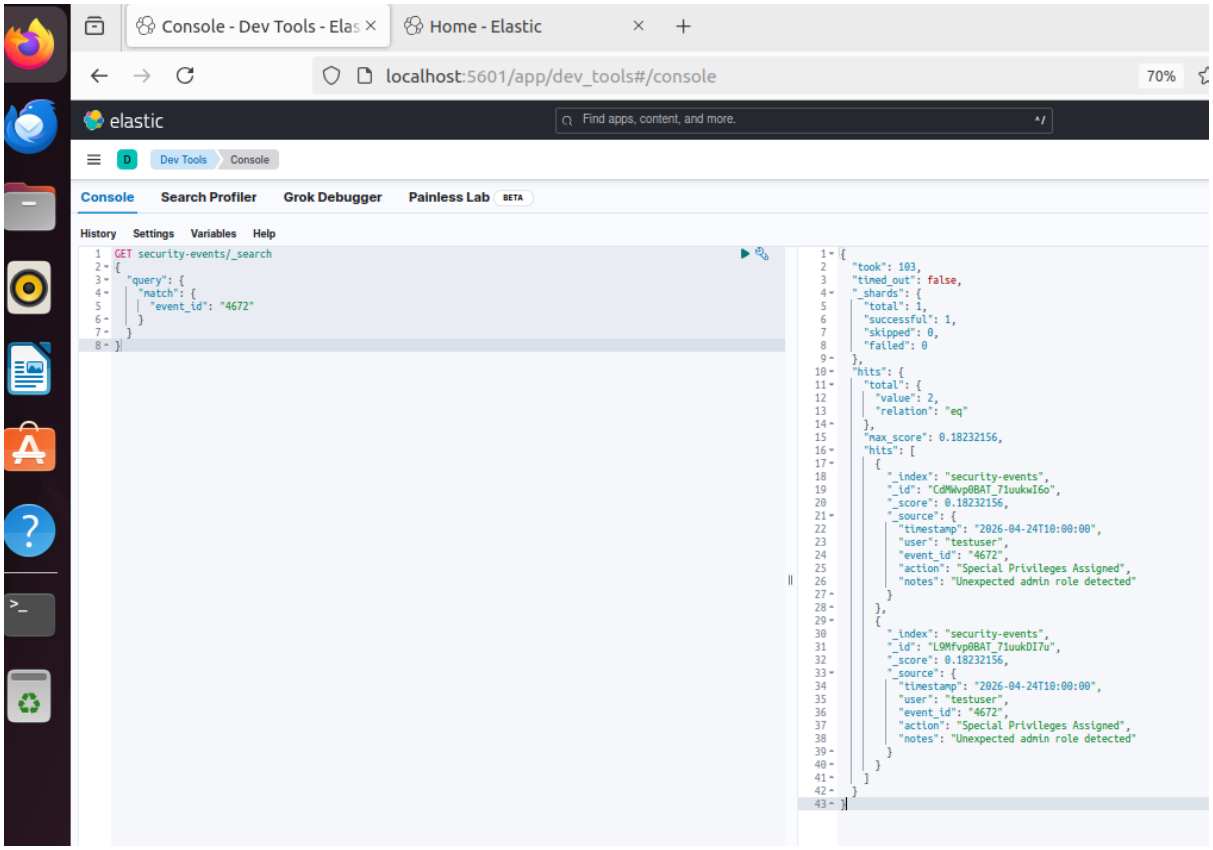
Subscribe (174)


Discover - Elastic
Home - Elastic
localhost:5601/app/discover#/?_g=(filters:!(),refreshInterval:(pause:!t,value:60000),time:(
elastic
Find apps, content, and more.
Discover
New
Open
Share
Alerts
Inspect
Save
security-events
event_id: 4672
Apr 24, 2026 @ 15:30:00.000
Apr 24, 2026 @ 15:30:00.000
Refresh
Search field names
Available fields
Empty fields
Meta fields
_id
_index
_score
2 hits
Break down by
Select field
Documents
Field statistics
Get the best look at your search results
Add relevant fields, reorder and sort columns, resize rows, and more in the document table.
Take the tour
Dismiss
timestamp
Document
Add a field


Console - Dev Tools - Elas
Home - Elastic
localhost:5601/app/dev_tools#/console
70%
elastic
Find apps, content, and more.
Dev Tools
Console
Console
Search Profiler
Grok Debugger
Painless Lab BETA
History
Settings
Variables
Help
GET security-events/_search
{
"query": {
"match": {
"event_id": "4672"
}
}
}
{
"took": 103,
"timed_out": false,
"_shards": {
"total": 1,
"successful": 1,
"skipped": 0,
"failed": 0
},
"hits": {
"total": {
"value": 2,
"relation": "eq"
},
"max_score": 0.18232156,
"hits": [
{
"_index": "security-events",
"_id": "CdMWvp0BAT_71uukwI6o",
"_score": 0.18232156,
"_source": {
"timestamp": "2026-04-24T10:00:00",
"user": "testuser",
"event_id": "4672",
"action": "Special Privileges Assigned",
"notes": "Unexpected admin role detected"
}
},
{
"_index": "security-events",
"_id": "L9Mfvp0BAT_71uukDI7u",
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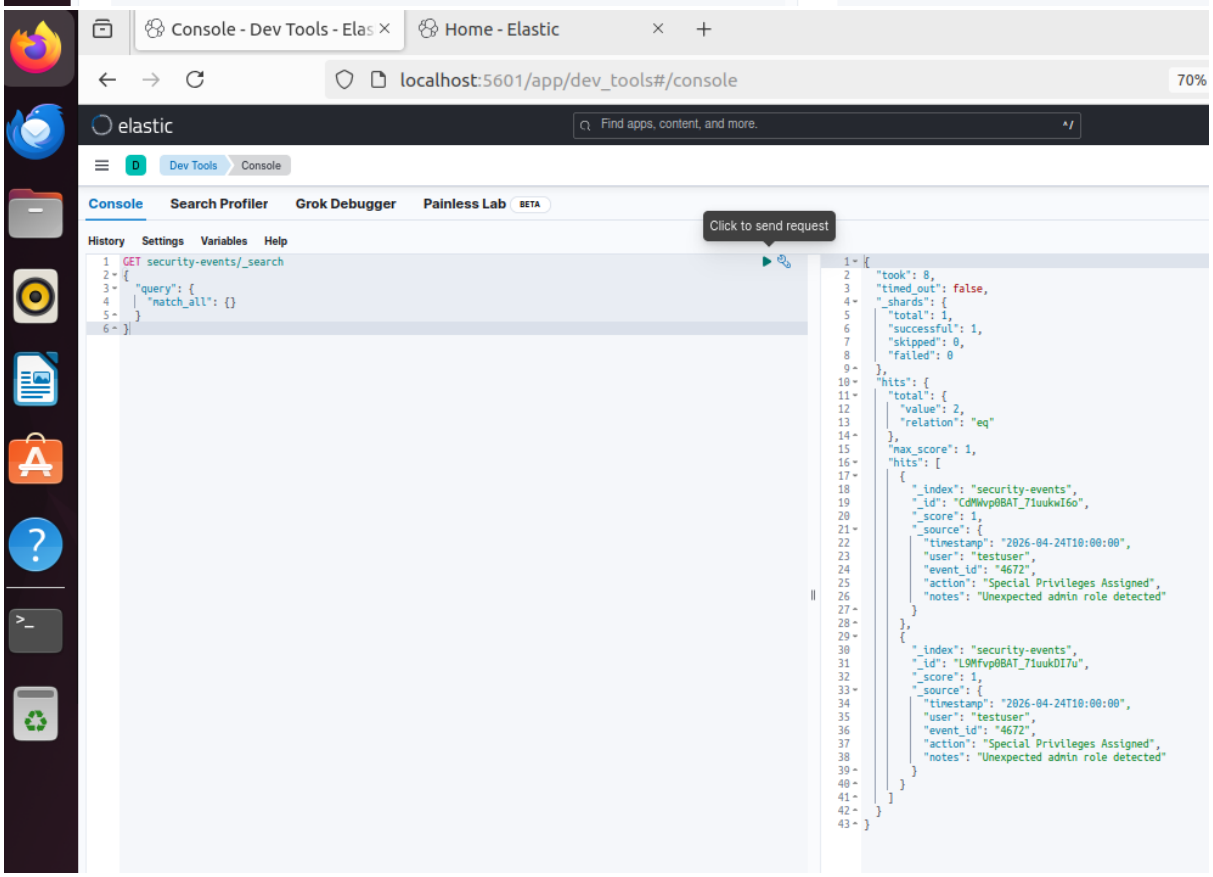
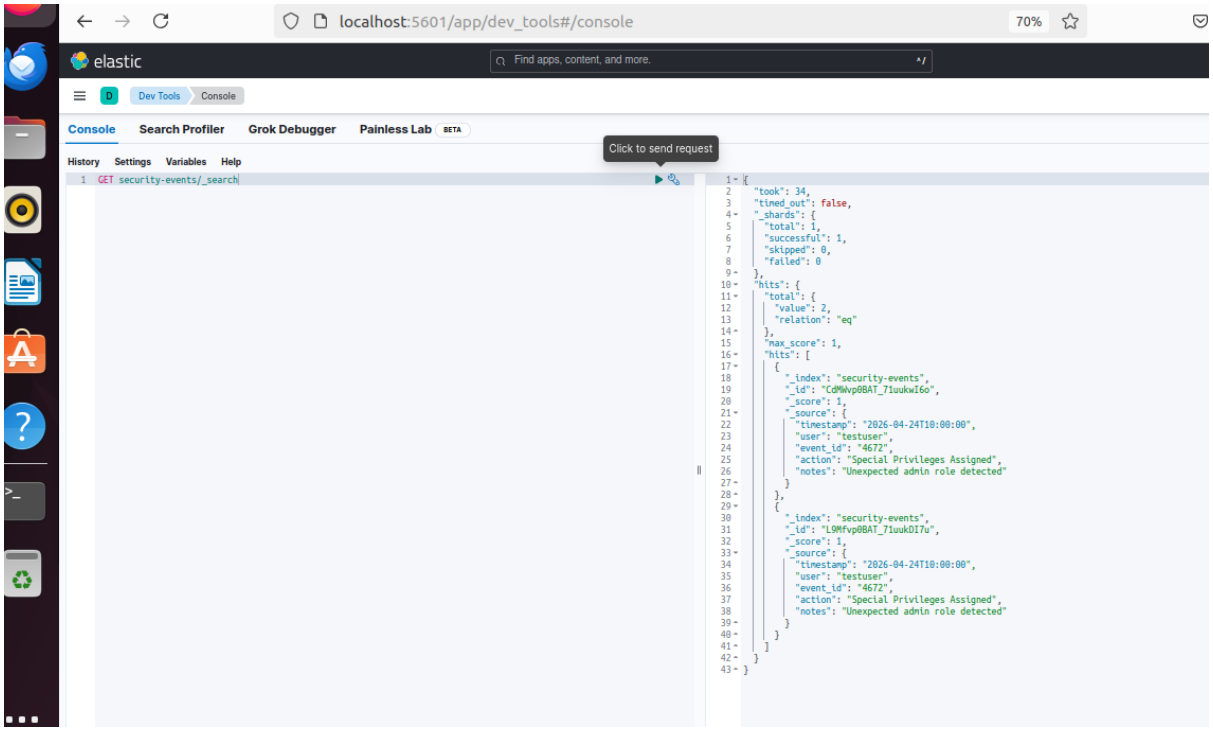
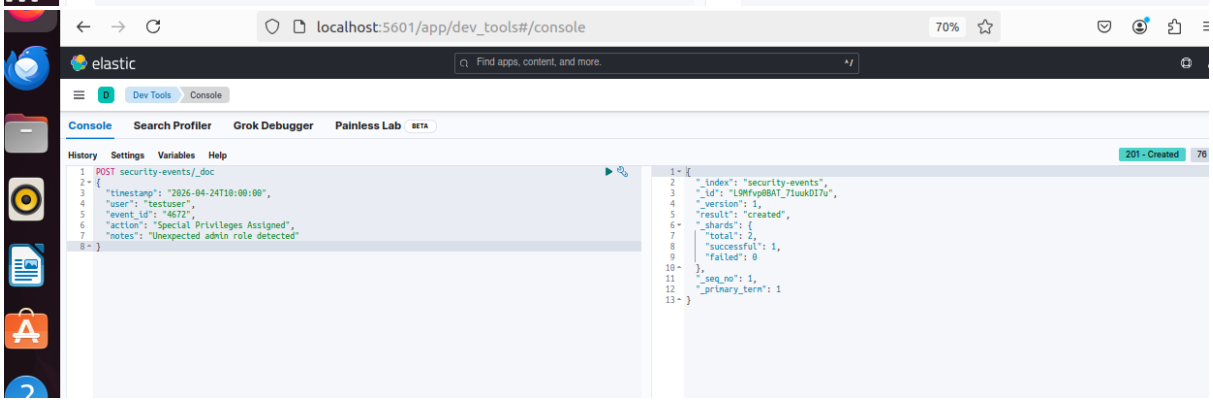
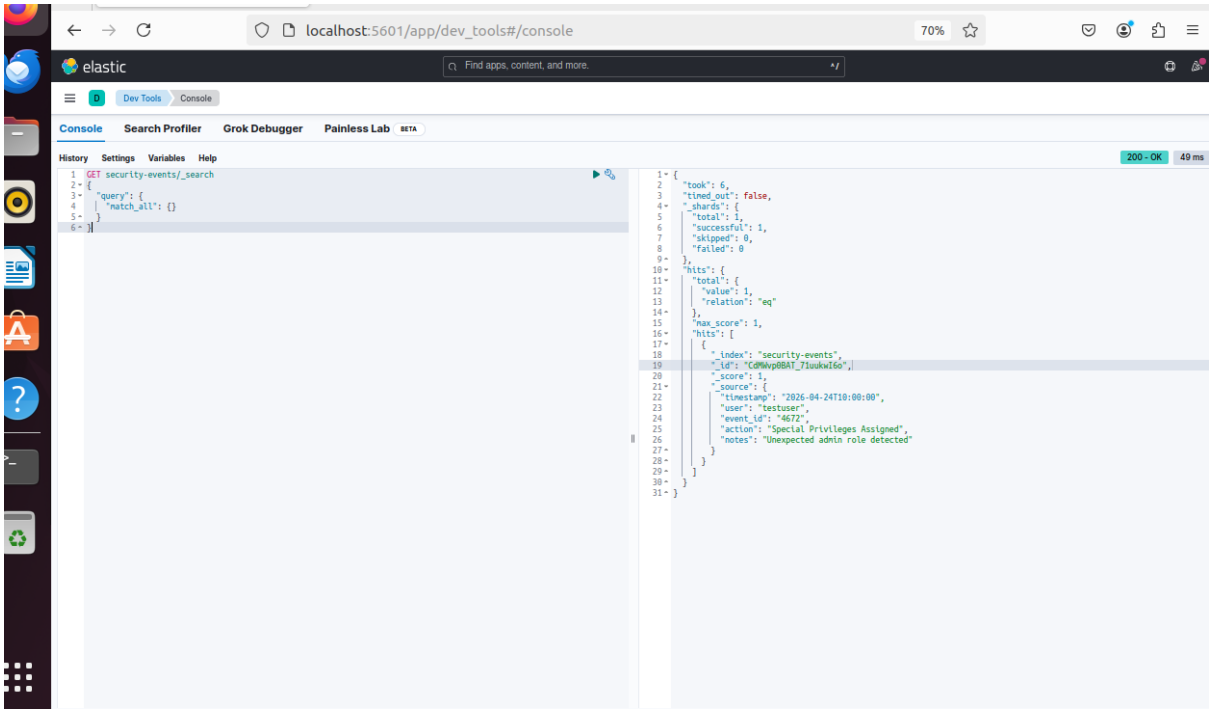
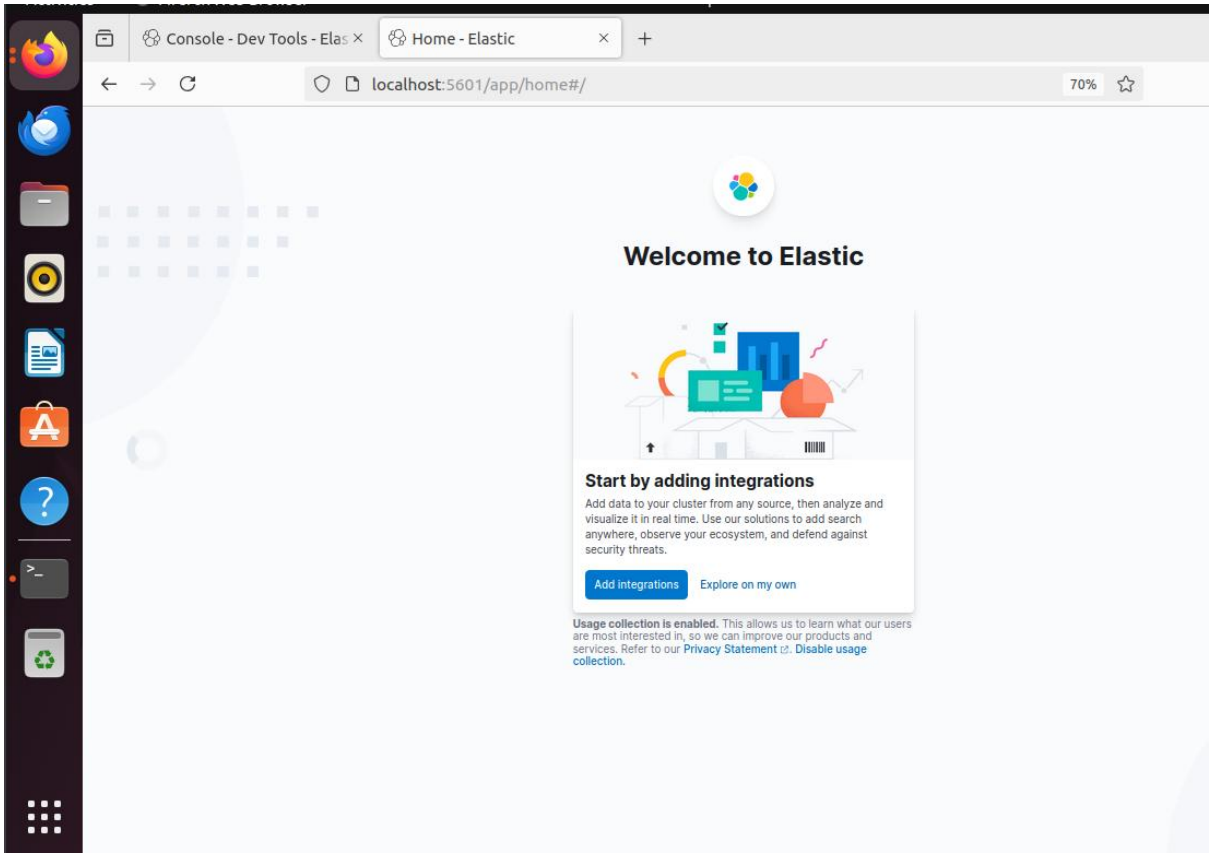
"_source": {
"timestamp": "2026-04-24T10:00:00",
"user": "testuser",
"event_id": "4672",
"action": "Special Privileges Assigned",
"notes": "Unexpected admin role detected"
}
}
]
}
}

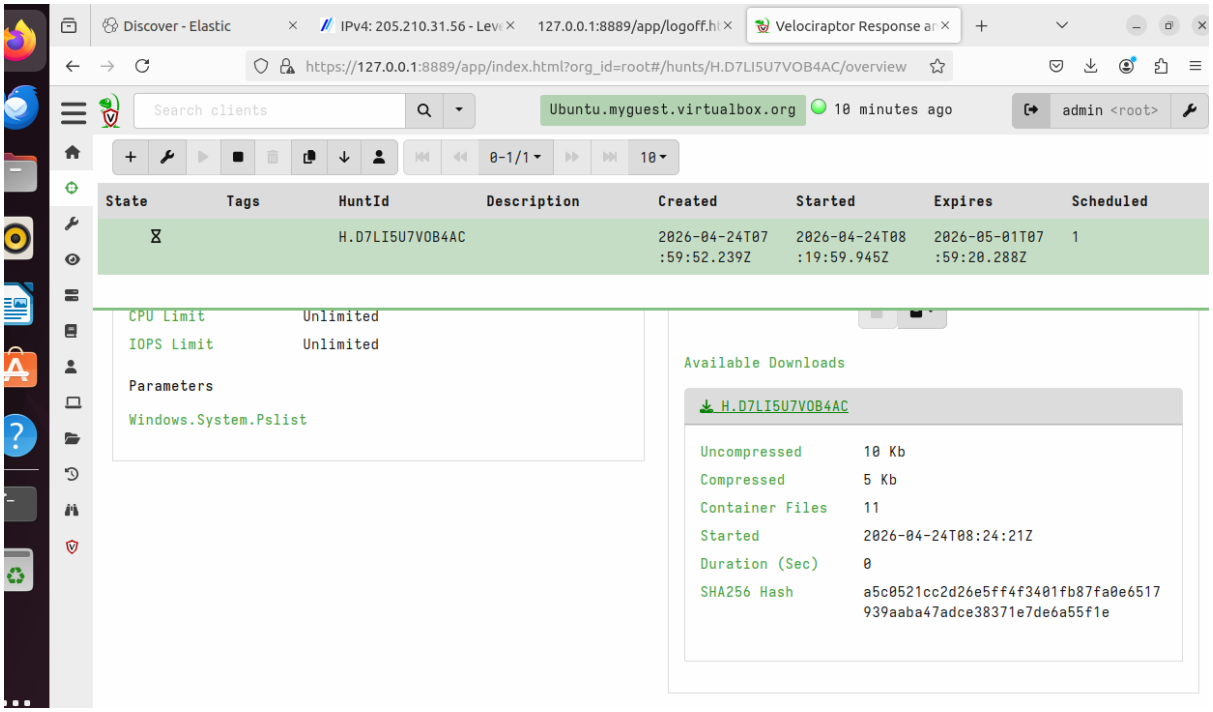

Console - Dev Tools - Elas
Home - Elastic
localhost:5601/app/dev_tools#/console
70%
elastic
Find apps, content, and more.
Dev Tools
Console
Console
Search Profiler
Grok Debugger
Painless Lab BETA
Click to send request
History
Settings
Variables
Help
GET security-events/_search
{
"query": {
"match_all": {}
}
}
{
"took": 8,
"timed_out": false,
"_shards": {
"total": 1,
"successful": 1,
"skipped": 0,
"failed": 0
},
"hits": {
"total": {
"value": 2,
"relation": "eq"
},
"max_score": 1,
"hits": [
{
"_index": "security-events",
"_id": "CdMWvp0BAT_71uukwI6o",
"_score": 1,
"_source": {
"timestamp": "2026-04-24T10:00:00",
"user": "testuser",
"event_id": "4672",
"action": "Special Privileges Assigned",
"notes": "Unexpected admin role detected"
}
},
{
"_index": "security-events",
"_id": "L9Mfvp0BAT_71uukDI7u",
"_score": 1,
"_source": {
"timestamp": "2026-04-24T10:00:00",
"user": "testuser",
"event_id": "4672",
"action": "Special Privileges Assigned",
"notes": "Unexpected admin role detected"
}
}
]
}
}


localhost:5601/app/dev_tools#/console
70%
elastic
Find apps, content, and more.
Dev Tools
Console
Console
Search Profiler
Grok Debugger
Painless Lab
BETA
History
Settings
Variables
Help
Click to send request
GET security-events/_search
{
"took": 34,
"timed_out": false,
"_shards": {
"total": 1,
"successful": 1,
"skipped": 0,
"failed": 0
},
"hits": {
"total": {
"value": 2,
"relation": "eq"
},
"max_score": 1,
"hits": [
{
"_index": "security-events",
"_id": "CdMWvp0BAT_7IuukwI6o",
"_score": 1,
"_source": {
"timestamp": "2026-04-24T10:00:00",
"user": "testuser",
"event_id": "4672",
"action": "Special Privileges Assigned",
"notes": "Unexpected admin role detected"
}
},
{
"_index": "security-events",
"_id": "L9MfvpBAT_7IuukDI7u",
"_score": 1,
"_source": {
"timestamp": "2026-04-24T10:00:00",
"user": "testuser",
"event_id": "4672",
"action": "Special Privileges Assigned",
"notes": "Unexpected admin role detected"
}
}
]
}
}


localhost:5601/app/dev_tools#/console
70%
elastic
Find apps, content, and more.
Dev Tools
Console
Console
Search Profiler
Grok Debugger
Painless Lab
BETA
History
Settings
Variables
Help
201 - Created
POST security-events/_doc
{
"timestamp": "2026-04-24T10:00:00",
"user": "testuser",
"event_id": "4672",
"action": "Special Privileges Assigned",
"notes": "Unexpected admin role detected"
}
{
"_index": "security-events",
"_id": "L9MfvpBAT_7IuukDI7u",
"_version": 1,
"result": "created",
"_shards": {
"total": 2,
"successful": 1,
"failed": 0
},
"_seq_no": 1,
"_primary_term": 1
}


localhost:5601/app/dev_tools#/console
70%
elastic
Find apps, content, and more.
Dev Tools
Console
Console
Search Profiler
Grok Debugger
Painless Lab
BETA
History
Settings
Variables
Help
200 - OK
49 ms
GET security-events/_search
{
"query": {
"match_all": {}
}
}
{
"took": 6,
"timed_out": false,
"_shards": {
"total": 1,
"successful": 1,
"skipped": 0,
"failed": 0
},
"hits": {
"total": {
"value": 1,
"relation": "eq"
},
"max_score": 1,
"hits": [
{
"_index": "security-events",
"_id": "CdMWvp0BAT_7luukwI6o",
"_score": 1,
"_source": {
"timestamp": "2026-04-24T10:00:00",
"user": "testuser",
"event_id": "4672",
"action": "Special Privileges Assigned",
"notes": "Unexpected admin role detected"
}
}
]
}
}


Console - Dev Tools - Elas
Home - Elastic
localhost:5601/app/home#/
70%
Welcome to Elastic
Start by adding integrations
Add data to your cluster from any source, then analyze and visualize it in real time. Use our solutions to add search anywhere, observe your ecosystem, and defend against security threats.
Add integrations
Explore on my own
Usage collection is enabled. This allows us to learn what our users are most interested in, so we can improve our products and services. Refer to our Privacy Statement. Disable usage collection.


Discover - Elastic
IPv4: 205.210.31.56 - Leve
127.0.0.1:8889/app/logoff.hi
Velociraptor Response ar
https://127.0.0.1:8889/app/index.html?org_id=root#/hunts/H.D7LI5U7VOB4AC/overview
Search clients
Ubuntu.myguest.virtualbox.org
10 minutes ago
admin <root>
0-1/1
10
State
Tags
HuntId
Description
Created
Started
Expires
Scheduled
H.D7LI5U7VOB4AC
2026-04-24T07:59:52.239Z
2026-04-24T08:19:59.945Z
2026-05-01T07:59:20.288Z
1
CPU Limit
Unlimited
IOPS Limit
Unlimited
Parameters
Windows.System.Pslist
Available Downloads
H.D7LI5U7VOB4AC
Uncompressed
10 Kb
Compressed
5 Kb
Container Files
11
Started
2026-04-24T08:24:21Z
Duration (Sec)
0
SHA256 Hash
a5c0521cc2d26e5ff4f3401fb87fa0e6517939aaba47adce38371e7de6a55f1e